

Network Asset Discovery Using Kali Linux

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1. Executive Summary

This project aimed to identify all active devices connected to the local home network. The objective was to build a basic asset inventory by discovering hosts and verifying their presence on the network.

The assessment was conducted using Kali Linux with network discovery tools such as Netdiscover and Nmap. The results revealed five active hosts on the network, including the router, personal computing devices, and consumer electronics.

Understanding the devices present on a network is an essential step in cybersecurity because unknown or unauthorized devices may indicate potential security risks.

2. Scope of Assessment

Assessment Type: Internal Network Asset Discovery

Target Network: 192.168.0.0/24

Testing Platform: Kali Linux Virtual Machine

Tools Used: Netdiscover, Nmap

Limitations:

- Testing was limited to the local home network environment
- Only host discovery and service verification were performed
- No exploitation or intrusive testing were conducted

3. Methodology

The asset discovery process followed a structured approach consisting of the following phases:

1. Network Interface Identification
2. ARP-Based Host Discovery
3. Network Host Verification

4. Basic Service Identification

4. Environment Setup

The testing environment consisted of a Kali Linux virtual machine running within a laptop using virtualization software. The virtual machine was configured to connect directly to the local network.

This setup allowed the Kali system to behave as an independent device on the network with its own IP address.

Network communication was verified using the primary network interface:

eth0

5. ARP-Based Network Discovery

Tool Used: Netdiscover

Command Executed: `sudo netdiscover -i eth0`

Command Explanation:

sudo	Executes the command with administrative privileges
netdiscover	ARP scanning tool used to identify devices on a network
-i eth0	Specifies the network interface to monitor

Result:

The tool captured ARP requests and responses from devices on the network and identified the following hosts:

IP Address	MAC Address	Vendor
192.168.0.4	b0:1c:0c:c7:6a:7d	Shenzhen Chuangwei-RGB Electronics
192.168.0.24	4c:49:6c:97:66:d4	Intel Corporation
192.168.0.1	a0:a3:f0:a8:d9:6c	D-Link International

These devices represent active hosts responding to ARP requests on the local network.

Currently scanning: 192.168.0.0/16 Screen View: Unique Hosts					
14 Captured ARP Req/Rep packets, from 3 hosts. Total size: 840					
IP	At MAC Address	Count	Len	MAC Vendor / Hostname	
192.168.0.4	b0:1c:0c:c7:6a:7d	1	60	SHENZHEN CHUANGWEI-RGB ELECTRONICS CO.,LTD	
192.168.0.24	4c:49:6c:97:66:d4	6	360	Intel Corporate	
192.168.0.1	a0:a3:f0:a8:d9:6c	7	420	D-Link International	

6. Network Host Verification

Tool Used: Nmap

Command Executed: nmap -sn 192.168.0.0/24

Command Explanation:

nmap	Network scanning tool
sn	Host discovery scan (no port scanning)
192.168.0.0/24	Target network range

Result:

The scan identified five active hosts within the network:

192.168.0.1	Router Gateway (D-Link)
192.168.0.4	Smart Device / IoT Device (Chuangwei Electronics)
192.168.0.6	Samsung Galaxy Tab A7 Lite Tablet
192.168.0.24	Laptop / PC (Intel)
192.168.0.27	Kali Linux Virtual Machine

This confirmed the presence of multiple devices connected to the network, including computing systems and consumer electronics.

(kali@kali)-[~]
\$ nmap -sn 192.168.0.0/24
Starting Nmap 7.95 (https://nmap.org) at 2026-03-11 20:14 EDT
Nmap scan report for RTK_GW.realtek (192.168.0.1)
Host is up (0.0083s latency).
MAC Address: A0:A3:F0:A8:D9:6C (D-Link International)
Nmap scan report for 192.168.0.4
Host is up (0.11s latency).
MAC Address: B0:1C:0C:C7:6A:7D (Shenzhen Chuangwei-RGB Electronics)
Nmap scan report for Galaxy-Tab-A7-Lite.realtek (192.168.0.6)
Host is up (0.11s latency).
MAC Address: 3E:85:F8:EA:27:D2 (Unknown)
Nmap scan report for HBANBAMBABRK.realtek (192.168.0.24)
Host is up (0.00021s latency).
MAC Address: 4C:49:6C:97:66:D4 (Intel Corporate)
Nmap scan report for kali.realtek (192.168.0.27)
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 115.15 seconds

7. Service Verification

To further verify network services, the router was scanned using Nmap.

Command Executed: nmap 192.168.0.1

Result:

Port 21	FTP	Filtered
Port 22	SSH	Filtered
Port 23	Telnet	Filtered
Port 53	DNS	Filtered
Port 80	HTTP	Open
Port 443	HTTPS	Open
Port 515	Printer	Filtered

The router exposes HTTP and HTTPS management interfaces while other administrative services are filtered by the firewall.

```
(kali㉿kali)-[~]
$ nmap 192.168.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-11 20:27 EDT
Nmap scan report for RTK_GW.realtek (192.168.0.1)
Host is up (0.0039s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    filtered  ftp
22/tcp    filtered  ssh
23/tcp    filtered  telnet
53/tcp    filtered  domain
80/tcp    open      http
443/tcp   open      https
515/tcp   filtered  printer
MAC Address: A0:A3:F0:A8:D9:6C (D-Link International)

Nmap done: 1 IP address (1 host up) scanned in 2.85 seconds
```

8. Findings

The network asset discovery process identified five active devices connected to the local network.

These include:

- Router gateway (192.168.0.1)
- Smart device/IoT equipment (192.168.0.4)
- Samsung tablet (192.168.0.6)
- Laptop/PC (192.168.0.24)
- Kali Linux virtual machine (192.168.0.27)

The router provides web-based management services through ports 80 (HTTP) and 443 (HTTPS), while other administrative services appear to be filtered by the router firewall.

No unauthorized or suspicious devices were identified during the assessment.

9. Conclusion

This project demonstrated the process of discovering and identifying devices within a local network environment.

Using Kali Linux and common network security tools, it was possible to enumerate active hosts and build a basic network asset inventory.

Asset discovery is an important step in cybersecurity operations because organizations must first understand what devices exist on their network before implementing effective security controls.